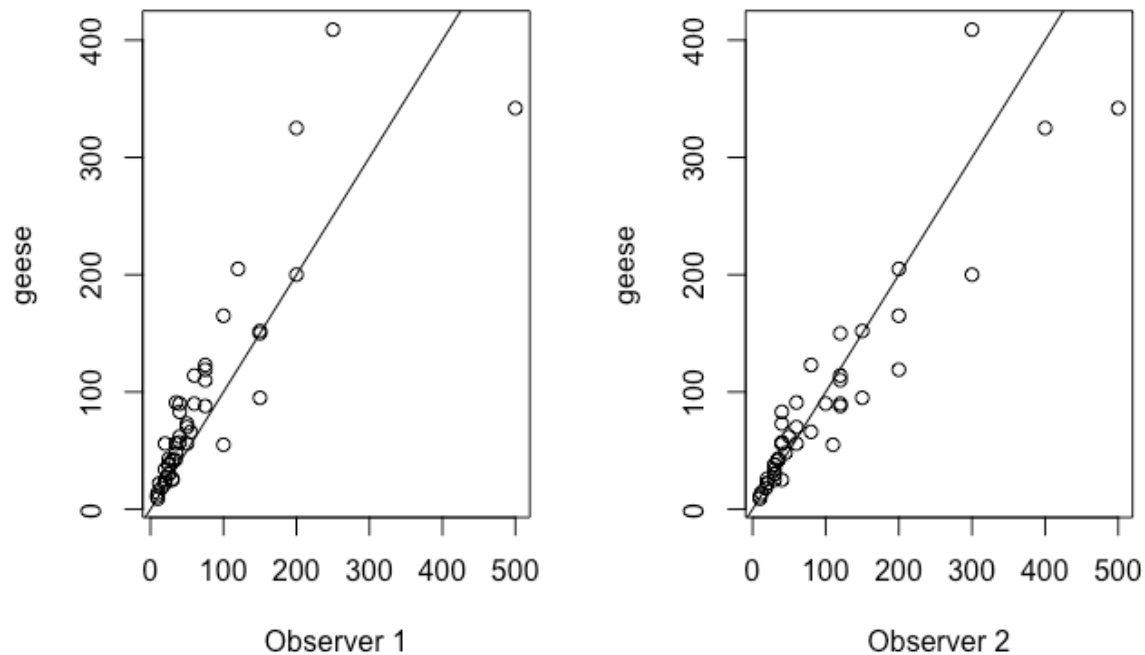


Assignment 7

Exercise 7.1

Part a.

Looking at the plots we see that there is a clear linear correlation in both plots.



Part b.

The function *lm* gives linear model

$$Lm1 = 26.6496 + obs1 * 0.8826 + error$$

$$Lm2 = 16.1643 + obs2 * 0.7691 + error$$

Consider test

$$H_0: B = 0$$

$$H_0: B \neq 0$$

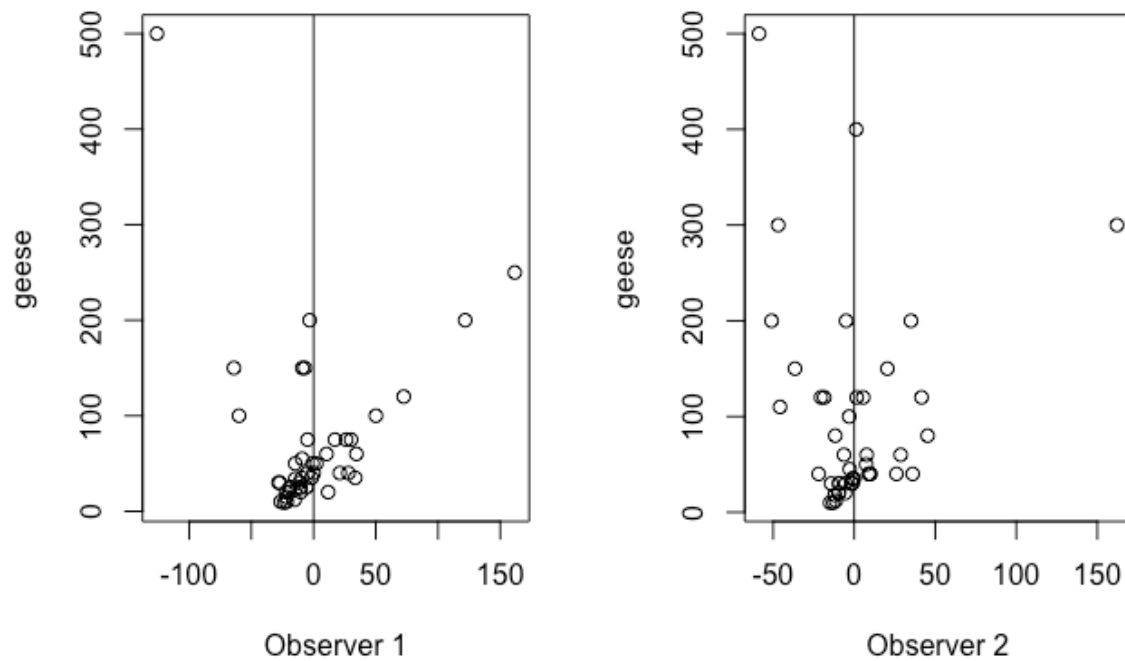
testing that both linear models are linear with the significance level 0.05 and the test statistic

$$Test = \frac{(45-1-1)(SSY-RSS)}{RSS}$$

$$P1 = 1.537e-14 \text{ and } P2 < 2.2 \text{ e-}16$$

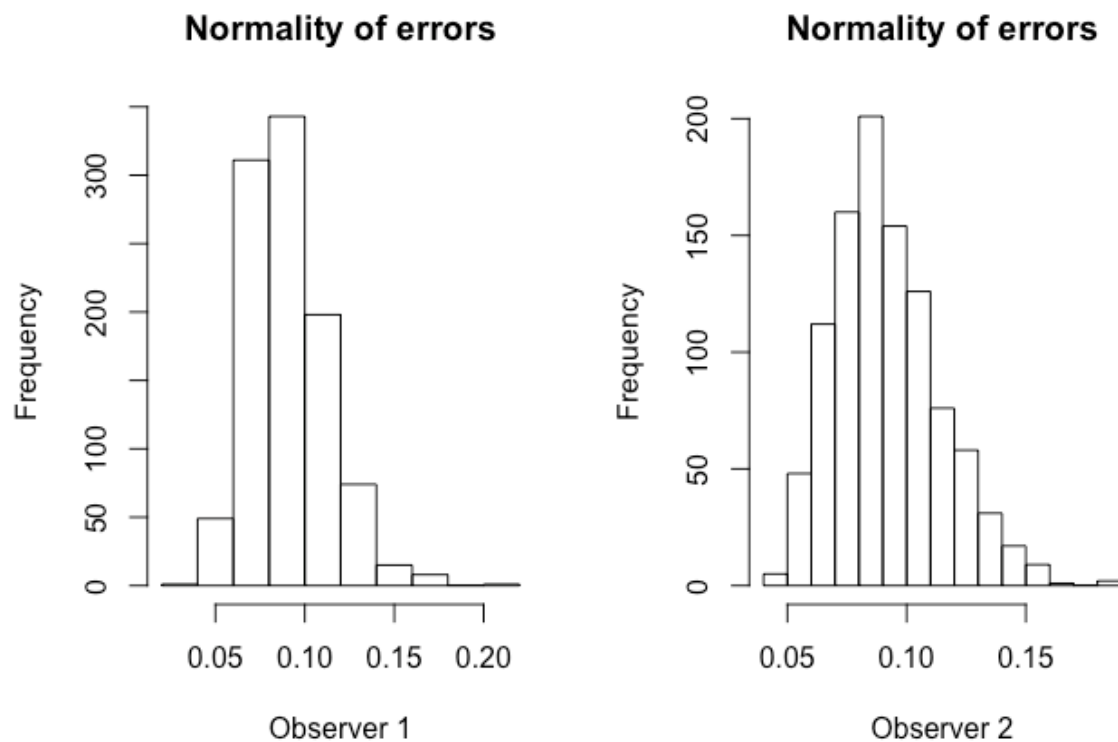
The p-values are extremely small, hence, the null hypothesis is rejected.

Part c.



The residuals from Observer 1 show that the most outliers are in the upper left corner. This is not so extreme that it proves that it is not linear. The residuals from Observer 2 show that it is not perfect linear but has less outliers.

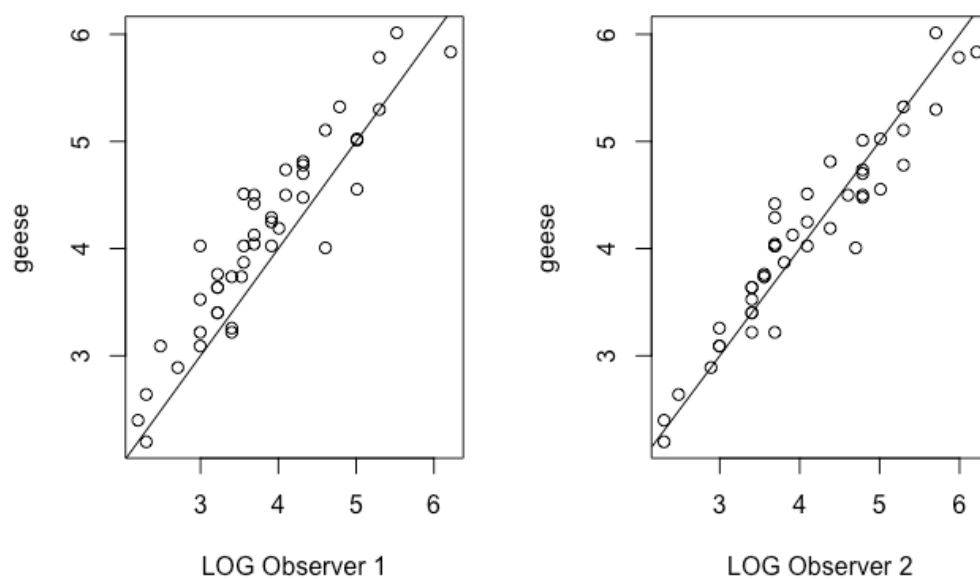
Part d.

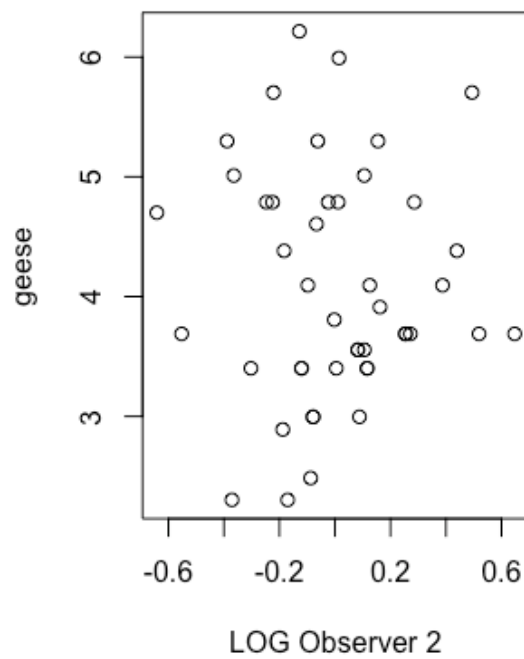
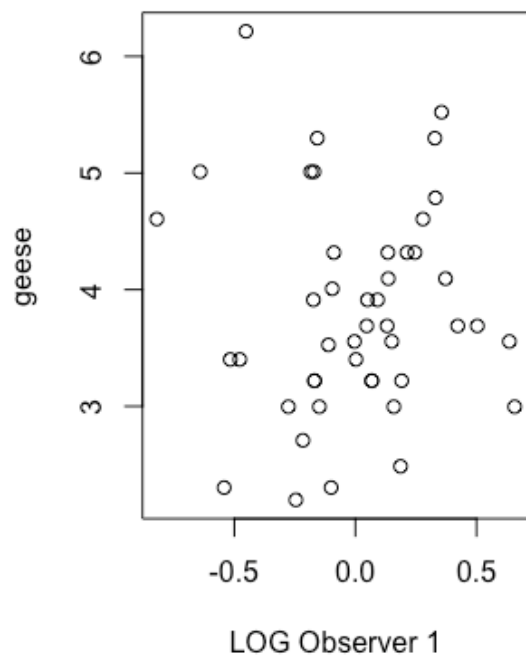


The *lm.norm.test*, *ks.test*(0.19419, 0.20399) and show that both are in the 100% quantile. There is a low p-value. So, the error on the linear model are small.

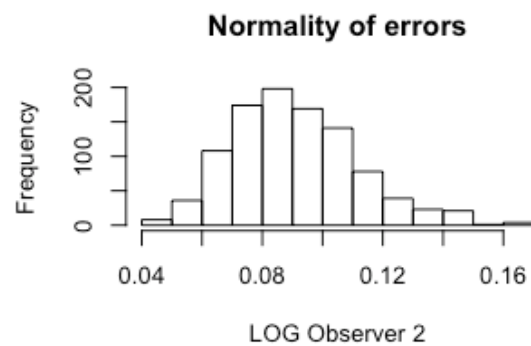
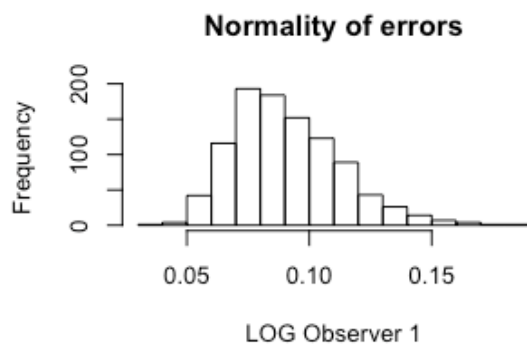
Part e.

The data show that it still linear.



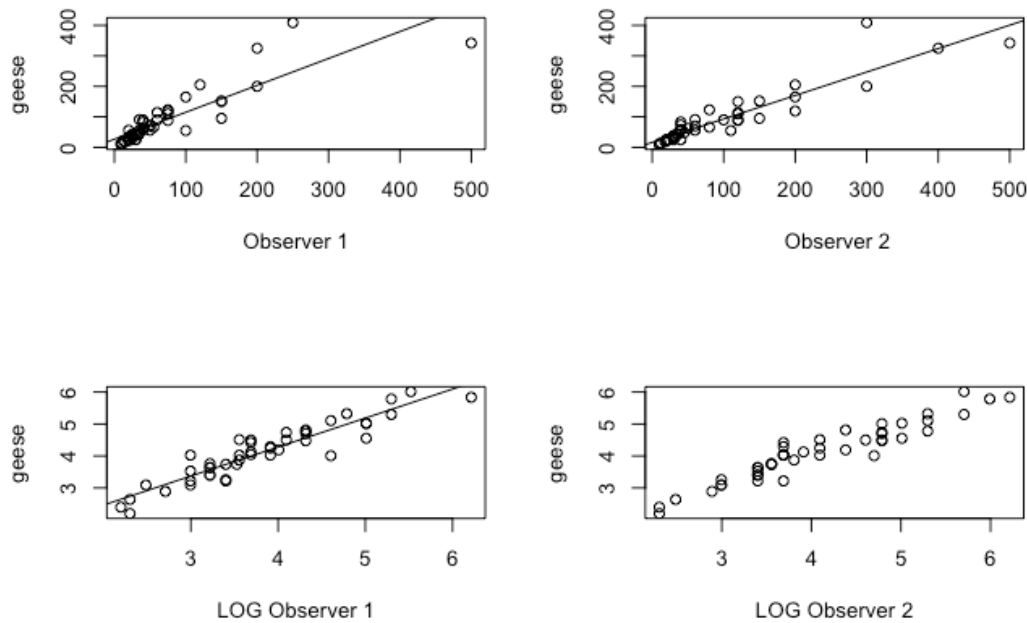


The Log observer shows that the residuals of the log observer still a small trend line.



Part f.

All plots show linear regressions. As expected, the Log observer have a better spread than the base version.



Part g.

The data show that both observers count the number of geese well. It is clearly linearly related. The Log of the data can also use for the observe.

Exercise 7.2

Part b.

The function composition *summary(lm)* gives four linear regression models

$$\text{Oxidant} = -1.43279 + 0.22993 \text{ Insolation} + \text{Error},$$

$$\text{Oxidant} = 27.4446 - 0.2088 \text{ Humidity} + \text{Error},$$

$$\text{Oxidant} = -46.4292 + 0.7850 \text{ Temperature} + \text{Error},$$

$$\text{Oxidant} = 45.3171 - 0.6331 \text{ Wind} + \text{Error}$$

with determination coefficients 0.2552, 0.124, 0.576, and 0.5863, respectively. Consider *t*-test with

$$H_0: \beta_k = 0; \beta_j \text{ arbitrary for } j \neq k,$$

$$H_1: \beta_k \neq 0; \beta_j \text{ arbitrary for } j \neq k,$$

and the test statistic

$$T_k = \frac{\hat{\beta}_k}{\sqrt{\widehat{\text{Cov}}(\hat{\beta})_{kk}}}$$

which follows a t_{n-p-1} -distribution under H_0 . The resulting model includes Wind because such model has the greatest determination coefficient 0.5863 and the corresponding p-value is less than the given significance level 0.05. Repeating this procedure gives three linear regression models

$$\text{Oxidant} = 32.32615 - 0.55639 \text{ Wind} + 0.13161 \text{ Insolation} + \text{Error},$$

$$\text{Oxidant} = 46.91570 - 0.60955 \text{ Wind} - 0.04516 \text{ Humidity} + \text{Error},$$

$$\text{Oxidant} = -5.20334 - 0.42706 \text{ Wind} + 0.52035 \text{ Temperature} + \text{Error}$$

with the greatest determination coefficient 0.7773 and the corresponding p-value less than 0.05 for the last model. Hence, the resulting model includes Wind and Temperature. The model does not consist of variables Wind, Temperature, and Humidity or Wind, Temperature, and Insolation because the corresponding p-values are greater than 0.05. Therefore, the resulting linear regression model is

$$\text{Oxidant} = -5.20334 - 0.42706 \text{ Wind} + 0.52035 \text{ Temperature} + \text{Error}.$$

Part c.

Consider Overall F -test with

$$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0,$$

$$H_1: \beta_j \neq 0 \text{ for some } j,$$

and the test statistic

$$F = \frac{25(SSY - RSS)}{4 \text{ RSS}}$$

which follows $F_{4,25}$ under H_0 . The composition function *summary(lm)* gives parameter estimates

$$\begin{aligned} \text{Oxidant} = & -15.4937 - 0.44291 \text{ Wind} + 0.56933 \text{ Temperature} + 0.09292 \text{ Humidity} \\ & + 0.02275 \text{ Insolation} + \text{Error} \end{aligned}$$

and a p-value 2.279e-08, which is less than 0.05. Hence, the test rejects H_0 and at least one of the variables should be included in the model.

Part d.

Use the t -test from Part b. Variable Insolation has the greatest insignificant p-value. Hence, the model is reduced to

$$\text{Oxidant} = -16.607 - 0.446 \text{ Wind} + 0.602 \text{ Temperature} + 0.099 \text{ Humidity} + \text{Error}.$$

Then variable Humidity has the greatest insignificant p-value, hence, the model is reduced to

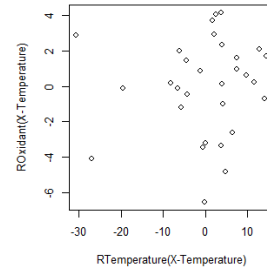
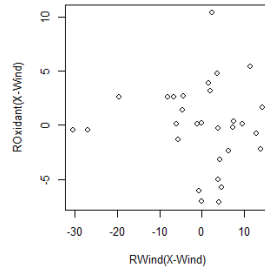
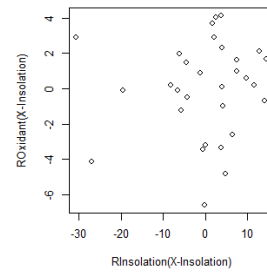
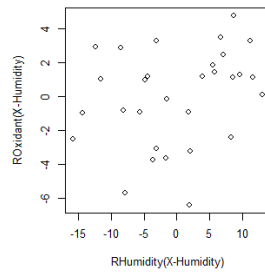
$$\text{Oxidant} = -5.20334 - 0.42706 \text{ Wind} + 0.52035 \text{ Temperature} + \text{Error}.$$

Part e.

Both obtained models are the same.

Part f.

There seems to be no relationship between Oxidant and any of the explanatory variables because there is no clear line or shape of data points in the added variable plots below. Since there is no apparent linear relationship between Oxidant and one of the explanatory variables, the model is not extended by any of the variables.



Part g.

Consider the significance test of δ for outliers with

$$H_0: \delta \leq 0,$$

$$H_1: \delta > 0$$

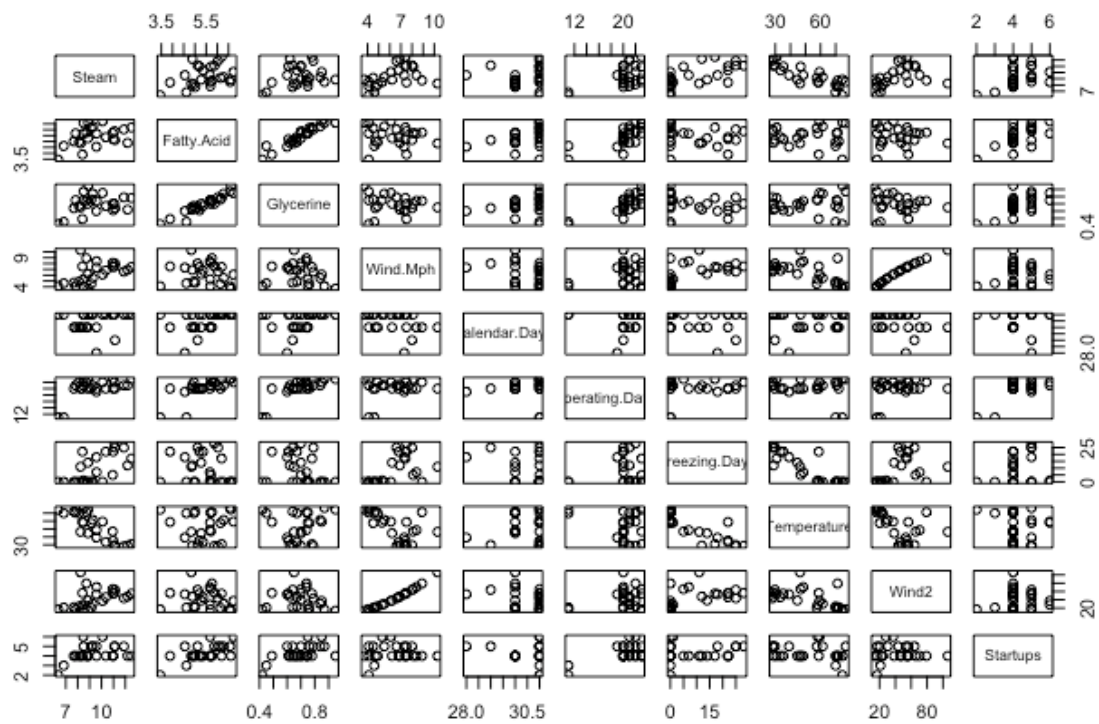
and the test statistic

$$T_{p+1} = \frac{\hat{\delta}}{\sqrt{\widehat{Cov}(\hat{\beta}, \hat{\delta})_{p+1, p+1}}}$$

which follows t_{n-p-2} under H_0 . The function `summary(lm)` gives the p-value 0.052 for the two-sided test. Hence, the p-value for the one-sided test is 0.026, which is lower than 0.05. Therefore, the test rejects H_0 , and the 4th data point is an outlier.

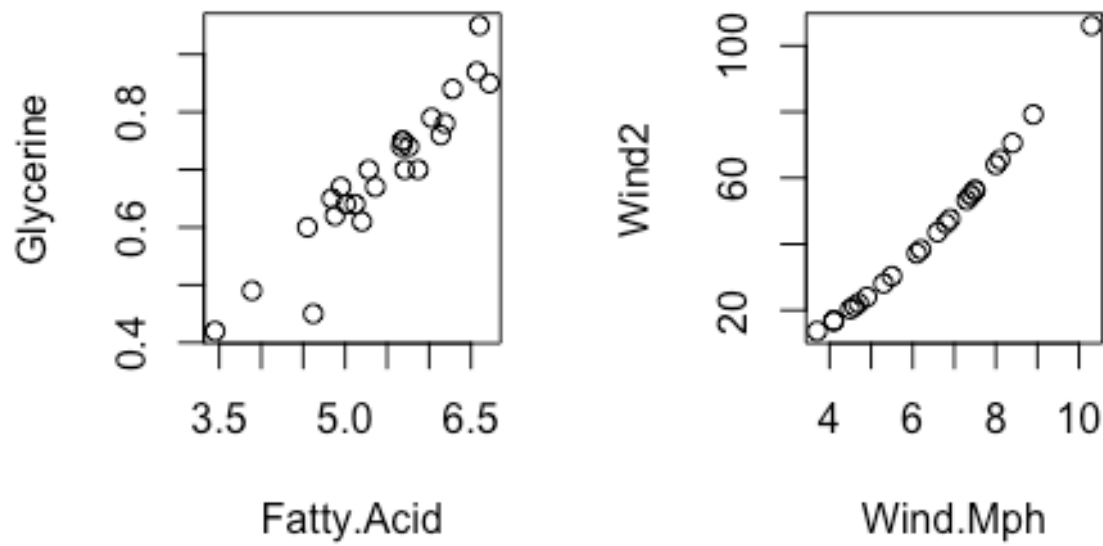
Exercise 7.3

Part a.



Look at the plot it shows to variable are clearly related fatty acid and Glycerine and Wind.Mph and Wind2. Looking at the variable and there relationship with steam there is trend with Temperature. This is confirmed by the data because it has the highest r.squared value.

Part b.



These plots show a clear line relationship between the variables.

Appendix

```
setwd("~/Documents/BBA3/SDA")
source('functions_Ch8.txt')
source('airpollution.txt')
source('geese.txt')
source('steamtable.txt')
```

#71a

```
geese = read.table("geese.txt", header = TRUE)
y=geese[,1]
x1=geese[,2]
x2=geese[,3]
```

```
par(mfrow=c(1,2))
plot(x1,y, ylab = "geese", xlab = "Observer 1")
abline(0,1)
plot(x2,y, ylab = "geese", xlab = "Observer 2")
abline(0,1)
```

#71b

```
lm1 = lm(y~x1)
lm2 = lm(y~x2)
summary.lm(lm1)
summary.lm(lm2)
lm1
lm2
```

#71c

```
par(mfrow=c(1,2))
plot(residuals(lm1),x1, xlab = "Observer 1", ylab = "geese")
abline(v=0)
plot(residuals(lm2),x2, xlab = "Observer 2", ylab = "geese")
abline(v=0)
```

#71d

```
hist(lm.norm.test(y,x1),xlab = "Observer 1", main = "Normality of errors")
hist(lm.norm.test(y,x2),xlab = "Observer 2", main = "Normality of errors")
ks1 = ks.test(residuals(lm1), 'pnorm', mean = mean(residuals(lm1)),sd = sd(residuals(lm1)))
ks1
ks2 = ks.test(residuals(lm2), 'pnorm', mean = mean(residuals(lm2)),sd = sd(residuals(lm2)))
ks2
```

#71e

```
loggeese = log(geese)
ly=loggeese[,1]
lx1=loggeese[,2]
lx2=loggeese[,3]
```

```

par(mfrow=c(1,2))
plot(lx1,ly, ylab = "geese", xlab = "LOG Observer 1")
abline(0,1)
plot(lx2,ly, ylab = "geese", xlab = "LOG Observer 2")
abline(0,1)
llm1 = lm(ly~lx1)
llm2 = lm(ly~lx2)
llm1
llm2
par(mfrow=c(1,2))
plot(residuals(llm1),lx1, xlab = "LOG Observer 1", ylab = "geese")
abline(h=0)
plot(residuals(llm2),lx2, xlab = "LOG Observer 2", ylab = "geese")
abline(h=0)
hist(lm.norm.test(ly,lx1),xlab = "LOG Observer 1", main = "Normality of errors")
hist(lm.norm.test(ly,lx2),xlab = "LOG Observer 2", main = "Normality of errors")
lks1 = ks.test(residuals(llm1), 'pnorm', mean = mean(residuals(llm1)),sd = sd(residuals(llm1)))
lks1
lks2 = ks.test(residuals(llm2), 'pnorm', mean = mean(residuals(llm2)),sd = sd(residuals(llm2)))
lks2
ecdf(lm.norm.test(ly,lx1))(lks1$statistic)
ecdf(lm.norm.test(ly,lx2))(lks2$statistic)

```

```

#71f
par(mfrow=c(2,2))
plot(x1,y, xlab = "Observer 1", ylab = "geese")
abline(lm1)
plot(x2,y, xlab = "Observer 2", ylab = "geese")
abline(lm2)
plot(lx1,ly, xlab = "LOG Observer 1", ylab = "geese")
abline(llm1)
plot(lx2,ly, xlab = "LOG Observer 2", ylab = "geese")
abline(llm2)

```

```

# Exercise 7.2
airpol = read.table("airpollution.txt", header = TRUE)
pairs(airpol[,c(6,2,3,4,5)])

```

```

# Part b.
Oxidant=airpol[,6]
Insolation=airpol[,5]
Humidity=airpol[,4]
Temperature=airpol[,3]
Wind=airpol[,2]

```

```

summary(lm(Oxidant~Insolation)) #0.2552
summary(lm(Oxidant~Humidity)) #0.124
summary(lm(Oxidant~Temperature)) #0.576

```

```
summary(lm(Oxidant~Wind)) #0.5863
```

```
summary(lm(Oxidant~Wind + Insolation)) #0.6613
```

```
summary(lm(Oxidant~Wind + Humidity)) #0.5913
```

```
summary(lm(Oxidant~Wind + Temperature)) #0.7773
```

```
summary(lm(Oxidant~Wind + Temperature + Insolation)) #0.7816
```

```
summary(lm(Oxidant~Wind + Temperature + Humidity)) #0.7964
```

```
# Part c.
```

```
summary(lm(Oxidant~Wind+Temperature+Humidity+Insolation))
```

```
# Part d.
```

```
summary(lm(Oxidant~Wind+Temperature+Humidity+Insolation))
```

```
summary(lm(Oxidant~Wind+Temperature+Humidity))
```

```
summary(lm(Oxidant~Wind+Temperature))
```

```
# Part f.
```

```
par(mfrow=c(2,2), pty="s")
```

```
ROxi_Hum = lm(Oxidant ~ Wind + Temperature + Insolation)$residuals
```

```
RHum_Hum = lm(Humidity ~ Wind + Temperature + Insolation)$residuals
```

```
plot(y=ROxi_Hum, x=RHum_Hum,
```

```
  ylab="ROxidant(X-Humidity)", xlab="RHumidity(X-Humidity)")
```

```
ROxi_Ins = lm(Oxidant ~ Wind + Temperature + Humidity)$residuals
```

```
RIns_Ins = lm(Insolation ~ Wind + Temperature + Humidity)$residuals
```

```
plot(y=ROxi_Ins, x=RIns_Ins,
```

```
  ylab="ROxidant(X-Insolation)", xlab="RInsolation(X-Insolation)")
```

```
ROxi_Wind = lm(Oxidant ~ Temperature + Humidity + Insolation)$residuals
```

```
RWind_Wind = lm(Wind ~ Temperature + Humidity + Insolation)$residuals
```

```
plot(y=ROxi_Wind, x=RIns_Ins,
```

```
  ylab="ROxidant(X-Wind)", xlab="RWind(X-Wind)")
```

```
ROxi_Tem = lm(Oxidant ~ Wind + Insolation + Humidity)$residuals
```

```
RTem_Tem = lm(Temperature ~ Wind + Insolation + Humidity)$residuals
```

```
plot(y=ROxi_Ins, x=RIns_Ins,
```

```
  ylab="ROxidant(X-Temperature)", xlab="RTemperature(X-Temperature)")
```

```
# Part g.
```

```
u = c(rep(0,3),1,rep(0,26))
```

```
msolm = lm(Oxidant ~ Wind + Temperature + u)
```

```
summary(msolm)
```

```
#73a
```

```
Steam = read.table('steamtable.txt', header = TRUE)
```

```
pairs(Steam)
```

```
Fatty_Acid = steamtable[,2]
Glycerine = steamtable[,3]
Wind_mph = steamtable[,4]
Calendar_Days = steamtable[,5]
Operating_Days = steamtable[,6]
Freezing_Days = steamtable[,7]
Temperature = steamtable[,8]
Wind2 = steamtable[,9]
Startups = steamtable[,10]
```

```
t(round(cor(Steam[,1],Steam[,-1]),3))
```

```
attach(Steam)
summary(lm(Steam ~ Fatty_Acid))$r.squared
summary(lm(Steam ~ Glycerine))$r.squared
summary(lm(Steam ~ Wind_Mph))$r.squared
summary(lm(Steam ~ Calendar_Days))$r.squared
summary(lm(Steam ~ Operating_Days))$r.squared
summary(lm(Steam ~ Freezing_Days))$r.squared
summary(lm(Steam ~ Temperature))$r.squared #(highest)
summary(lm(Steam ~ Wind2))$r.squared
summary(lm(Steam ~ Startups))$r.squared
```

```
#73b
```

```
par(mfrow=c(1,2))
plot(Fatty.Acid, Glycerine)
plot(Wind.Mph, Wind2)
```